SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 - SOFTWARE TESTING

EXPERIMENT 12

Railway Reservation System – Negative Test Case Study

Domain: Railway Reservation System

1. AIM

To identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases.

2. PROBLEM STATEMENT

The Railway Reservation System must reject missing journey details, past dates, unavailable trains, invalid passenger information, failed payments, and invalid cancellation requests.

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

S.No.	Requirement
1	Missing source/destination must be rejected.
2	Past journey dates must be rejected.
3	Unavailable trains/classes must not be booked.
4	Invalid passenger details must be rejected.
5	Failed payment must not confirm a ticket.
6	Cancellation must follow eligibility rules.

4. DEFECT REPORT

Defect ID	Defect / Scenario	Expected Result	Actual Result	Severity	Status
NT01	Search with missing station	Leave source/destination blank	Validation message displayed	Pass	
NT02	Select past journey date	Enter a past date	Date rejected	Pass	
NT03	Book unavailable class	Select class with no seats	Booking prevented	Pass	
NT04	Invalid passenger age	Enter invalid/negative age	Age rejected	Pass	
NT05	Blank passenger name	Leave mandatory name blank	Validation displayed	Pass	
NT06	Confirm after payment failure	Simulate declined payment	Ticket must not be confirmed	Fail	
NT07	Invalid payment details	Enter invalid payment data	Payment rejected	Pass	
NT08	Cancel non-eligible booking	Attempt invalid cancellation	Cancellation rejected	Pass	

5. RESULT

The Railway Reservation System requirements were analyzed and the negative test cases were prepared.

5. DETAILED EXECUTION OF A FAILED TEST CASE

Test Case NT06
Failed

Test scenario: Payment failure during booking

Precondition

Train and passenger details are valid, but payment is declined.

Test Steps

1. Search train.
2. Select available class.
3. Enter passenger details.
4. Simulate declined payment.
5. Observe booking status.

Expected Result

No ticket/PNR should be generated and booking should remain unconfirmed.

Actual Result

The system generates a booking confirmation even though payment failed.

Conclusion

Booking confirmation is not correctly linked to payment success.

6. SAMPLE BUG REPORT

Field	Details
Bug ID	BR-001
Linked Test Case	NT06
Severity	High
Priority	High
Bug Title	Railway system confirms a ticket after failed payment.
Environment	Application: Railway Reservation System; Browser: Google Chrome; Operating System: Windows 10
Preconditions	Train and passenger details are valid, but payment is declined.
Steps to Reproduce	1. Search train. 2. Select available class. 3. Enter passenger details. 4. Simulate declined payment. 5. Observe booking status.
Expected Result	No ticket/PNR should be generated and booking should remain unconfirmed.
Actual Result	The system generates a booking confirmation even though payment failed.
Impact	May cause incorrect booking/submission data, financial or academic processing errors, or poor user experience.
Suggested Fix	Validate the condition on both client and server sides and enforce the rule again at final submission.
Status	Open — awaiting developer investigation and correction.

7. BUG LIFE CYCLE

New — Tester reports the defect
Assigned — Developer receives the defect
In Progress — Developer investigates and fixes it
Fixed — Developer submits the correction
Retesting — Tester verifies the correction
Closed — Tester confirms the defect is resolved
If the defect still exists after the fix, the tester reopens the bug and sends it back for further correction.